SIMATS ENGINEERING

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CO1-AT2

Concept Mapping

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Machine learning

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Question 8: Candidate Elimination Algorithm for Customer Subscription Prediction

Introduction

The Candidate Elimination Algorithm is a supervised machine learning algorithm used to find all hypotheses that are consistent with the given training examples. It maintains two boundaries:

- **Specific Boundary (S):** The most specific hypothesis consistent with the positive examples.
- **General Boundary (G):** The most general hypothesis consistent with the training data.

The objective is to determine the version space for predicting whether a customer will subscribe to a service based on attributes such as Age Group, Income Level, Internet Usage, and Device Type.

Dataset

Example	Age Group	Income Level	Internet Usage	Device Type	Subscribe
1	Young	High	Frequent	Mobile	Yes
2	Young	High	Occasional	Mobile	Yes
3	Middle	Low	Frequent	Desktop	No
4	Young	Medium	Frequent	Mobile	Yes
5	Senior	High	Occasional	Desktop	No

Step 1: Initialize Boundaries

Specific Boundary:

$S0=\langle \emptyset,\emptyset,\emptyset,\emptyset \rangle$

General Boundary:

$G0=\langle ?,?,?,? \rangle$

Step 2: Example 1 (Positive)

(Young, High, Frequent, Mobile)

S_1 = ⟨Young, High, Frequent, Mobile⟩

G_1 = ⟨?, ?, ?, ?⟩

Step 3: Example 2 (Positive)

(Young, High, Occasional, Mobile)

Generalize Internet Usage.

S_2 = ⟨Young, High, ?, Mobile⟩

G_2 = ⟨?, ?, ?, ?⟩

Step 4: Example 3 (Negative)

(Middle, Low, Frequent, Desktop)

Specialize G to exclude the negative example.

G_3 = {
⟨Young, ?, ?, ?⟩,
⟨?, High, ?, ?⟩,
⟨?, ?, ?, Mobile⟩
}

S_3 unchanged.

Step 5: Example 4 (Positive)

(Young, Medium, Frequent, Mobile)

Generalize Income Level.

S_4 = ⟨Young, ?, ?, Mobile⟩

G_4 = {
⟨Young, ?, ?, ?⟩,
⟨?, ?, ?, Mobile⟩
}

Step 6: Example 5 (Negative)

(Senior, High, Occasional, Desktop)

No change required.

Final Boundaries

S = ⟨Young, ?, ?, Mobile⟩

G = {
⟨Young, ?, ?, ?⟩,
⟨?, ?, ?, Mobile⟩
}

Conclusion

The Candidate Elimination Algorithm learns that customers are likely to subscribe if they are Young and/or use a Mobile device. The final version space is represented by the above S and G boundaries.

Question 9: Candidate Elimination Algorithm for Healthy Lifestyle Prediction

<u>Introduction</u>

The Candidate Elimination Algorithm identifies all hypotheses consistent with the training data by maintaining the Specific Boundary (S) and General Boundary (G).

<u>Step 1: Initialize</u>

$\mathbf{S_0}$ = ⟨Ø, Ø, Ø, Ø⟩

$\mathbf{G_0}$ = ⟨?, ?, ?, ?⟩

<u>Step 2: Example 1 (Positive)</u>

(Balanced, Regular, Adequate, Low)

$\mathbf{S_1}$ = ⟨Balanced, Regular, Adequate, Low⟩

<u>Step 3: Example 2 (Positive)</u>

(Balanced, Regular, Poor, Medium)

Generalize Sleep and Stress Level.

$\mathbf{S_2}$ = ⟨Balanced, Regular, ?, ?⟩

<u>Step 4: Example 3 (Negative)</u>

(Junk, None, Poor, High)

Specialize G.

$\mathbf{G_3}$ = {
⟨Balanced, ?, ?, ?⟩,
⟨?, Regular, ?, ?⟩
}

<u>Step 5: Example 4 (Positive)</u>

(Balanced, Moderate, Adequate, Low)

Generalize Exercise.

$\mathbf{S_4}$ = ⟨Balanced, ?, ?, ?⟩

$\mathbf{G_4}$ = {
⟨Balanced, ?, ?, ?⟩
}

<u>Step 6: Example 5 (Negative)</u>

(Junk, Regular, Adequate, High)

No change required.

<u>Final Boundaries</u>

S = ⟨Balanced, ?, ?, ?⟩

G = {⟨Balanced, ?, ?, ?⟩}

<u>Conclusion</u>

The algorithm indicates that a Balanced Diet is the key factor associated with a healthy lifestyle in the given dataset.

Question 10: Machine Learning System for Disease Prediction

Introduction

Machine Learning can be used in healthcare to predict diseases such as diabetes and heart disease by analyzing patient medical records and lifestyle factors.

Proposed System

Input Features

- Age
- Blood Pressure
- Glucose Level
- Cholesterol
- BMI
- Lifestyle Habits

Algorithm Used

Random Forest Classifier

Reasons:

- High accuracy
- Handles large datasets
- Reduces overfitting
- Identifies important features

Steps Involved

1. Data Collection

Collect patient records from hospitals and healthcare databases.

2. Data Preprocessing

- Remove missing values
- Remove duplicates
- Normalize data
- Encode categorical attributes

3. Feature Selection

Select important features such as:

- Age
- Glucose Level
- Blood Pressure

- BMI

4. Model Training

- Split data into training and testing sets.
- Train Random Forest model.

5. Model Evaluation

Evaluate using:

- Accuracy
- Precision
- Recall
- F1-Score

Conclusion

A Random Forest-based disease prediction system can accurately predict diseases and assist doctors in early diagnosis and treatment planning.

Question 11: Architecture of a Typical Machine Learning Pipeline

Introduction

A Machine Learning pipeline is a sequence of steps used to develop, evaluate, and deploy a machine learning model.

Architecture

Data Collection
↓
Data Preprocessing
↓
Feature Engineering
↓
Model Training
↓
Model Evaluation
↓
Model Deployment

Stages

1. Data Collection

Gather data from databases, sensors, or surveys.

2. Data Preprocessing

Clean data by handling missing values and duplicates.

3. Feature Engineering

Create useful features and select relevant attributes.

4. Model Training

Train algorithms such as Decision Tree, SVM, or Random Forest.

5. Model Evaluation

Measure performance using Accuracy, Precision, Recall, and F1-Score.

6. Deployment

Deploy the model as a web application or cloud service.

Conclusion

A machine learning pipeline ensures systematic development, evaluation, and deployment of accurate models.

Question 12: Spam Email Classification

(a) Choice of Algorithm

Naïve Bayes Classifier

Reasons:

- Fast and efficient
- Suitable for text data
- High performance in email classification
- Requires less computational resources

<u>(b) Preprocessing Steps</u>

1. Tokenization

Split email text into words.

Example:

"Win a free prize"

→ [Win, Free, Prize]

2. Stop Word Removal

Remove common words such as "a", "the", "is".

3. Stemming/Lemmatization

Convert words to root forms.

Example:

winning → win

prizes → prize

4. Vectorization

Convert text into numerical form using:

- Bag of Words (BoW)
- TF-IDF

<u>(c) Evaluation Metrics</u>

Accuracy

Measures overall correctness of predictions.

Precision

Measures how many predicted spam emails are actually spam.

Recall

Measures how many actual spam emails are correctly identified.

F1-Score

Harmonic mean of Precision and Recall.

Conclusion

Using Naïve Bayes with proper preprocessing and evaluation metrics enables effective spam email classification with high accuracy.